Prasanth Rayudu

Graduate Engineer

rprasanth2024@gmail.com
(408) 396-1546
San Diego, CA
linkedin.com/in/prasanth-rayudu

SUMMARY

Computer Engineering M.S. candidate (Dec 2026) with hands-on experience in micro-architecture development and RTL design for datapath-heavy accelerators. Skilled in Verilog/SystemVerilog RTL implementation, architecture exploration, and soft/hard IP integration for SoC subsystems, with experience in ASIC/FPGA RTL design flows, synthesis, static timing analysis, and timing/power closure. Collaborated with verification and emulation teams on test-plan development and debug for pre-silicon validation; built Python/Tcl tooling for PPA trade-off and dynamic power estimation.

WORK EXPERIENCE

Graduate Research Assistant, Hardware-Software Co-Design for Edge AI Aug 2025 - Present
San Diego State University

- Developed HyperOPT, a Python-based hardware-software co-design framework for architecture-level design-space exploration of accelerator configurations under accuracy, area, latency, and energy constraints, enabling architecture exploration and PPA trade-offs for ASIC/FPGA accelerators.
- Modeled quantization, analog noise, and memory-cell non-idealities to quantify performance-per-watt and accuracy trade-offs, validating proxy estimates against detailed architecture-level results.
- Implemented pruning and binary-search algorithms to accelerate design-space exploration and produced reproducible benchmark reports for design reviews.

FPGA Emulation Flow Automation with CI/CD Integration Jan 2024 - Aug 2025
San Diego State University

- Integrated new CI/CD and FPGA tooling to build a scalable emulation flow with CI-triggered builds, regression execution, and test generation on Xilinx UltraScale platforms, supporting emulation-based test-plan development and pre-silicon validation.
- Developed Python, Tcl, Make, and Bash automation for compilation, environment configuration, and benchmark execution, integrating build and validation stages.
- Added log parsing and failure clustering enabling root-cause triage for over 95% of simulation-to-emulation mismatches, and reduced compile and debug iteration time by 30%.

Teaching Associate, General Studies Aug 2024 - May 2025
San Diego State University

- Led weekly first-year seminar sections, delivering instruction and timely assessment through Canvas LMS.
- Communicated complex technical material clearly to diverse audiences and collaborated in weekly teaching communities to improve course outcomes.

Student Intern, FPGA and Machine Learning Systems Jun 2021 - Jul 2022
In22 Labs

- Designed and validated Verilog RTL blocks for a biomedical machine-learning pipeline, co-simulating embedded C reference models to verify functional correctness.
- Automated ModelSim waveform and log validation in Python, reducing manual debug effort by 50% and improving regression reproducibility.
- Collaborated with FPGA, software, and data-science engineers to reduce end-to-end inference latency and support ARM-based hardware-software integration.

PROJECTS

Sep 2022 - Aug 2023

RTL Design and Verification of a Custom Execution Datapath

- Designed Verilog/SystemVerilog RTL for a custom execution datapath, including control logic and ALU-style math datapath for texture and math engines, evaluating micro-architectural trade-offs across area, timing, and power.
- Implemented synthesizable control FSMs and datapath blocks targeting logic synthesis and timing-driven implementation flows.
- Resolved clock-domain crossing issues and conducted static timing analysis to achieve timing closure on a Xilinx MPSoC FPGA prototype using Vivado.
- Collaborated with architecture and verification teams during design reviews to meet performance and power targets across compute workloads.

Aug 2022 - May 2023

FPGA-Based Neural Network Video Processor

- Designed a Verilog datapath for a real-time inference pipeline built from parallel MAC-based ALU units, bias-addition logic, control FSMs, and ROM-based activation.
- Achieved 30 FPS with over 95% classification accuracy through RTL simulation, timing analysis, and on-board debug using Vivado, ModelSim, and Xilinx ILA.
- Evaluated throughput, resource utilization, and power trade-offs to guide micro-architectural decisions.
- Automated build and test-generation flows with Tcl and Make to accelerate design iteration.

Jan 2023 - Jun 2024

System-Level FPGA Emulation Framework for SoC Subsystem Validation

- Integrated a multi-IP CPU-microcontroller subsystem on a Xilinx MPSoC FPGA for pre-silicon validation and ARM embedded-Linux bring-up.
- Created reusable SystemVerilog testbenches, BFMs, and transactors for AXI, interrupts, PCIe, I2C, and LVDS interfaces, supporting verification test-plan development and debug.
- Cut build cycles by 40% through reusable verification components and automation.
- Debugged system-level integration across CPU, memory, and peripheral interfaces during subsystem bring-up.

Jun 2023 - Present

Dynamic Power Profiling Framework for SoC RTL Validation

- Developed a Python tool to parse VCD activity, compute switching factors, and estimate dynamic power across multiple RTL execution and datapath blocks.
- Generated IP-level reports on clock gating and bus utilization to guide power optimization during design closure.
- Profiled power across datapath and control blocks to identify hotspots for micro-architectural improvement.
- Demonstrated 88% correlation of power estimates against post-silicon measurements.

SKILLS

Programming & Tooling
Python, C, C++, Tcl, Bash/Shell, Make, Git, Linux, CI/CD

Hardware & RTL
RTL design, Verilog, SystemVerilog, Datapath design, Micro-architecture, Synthesis, Static timing analysis, Timing closure, Clock-domain crossing, FPGA implementation, Vivado, Xilinx UltraScale/UltraScale+, MPSoC, VHDL, Xilinx ILA, ASIC RTL design

Verification & Validation
SystemVerilog testbenches, BFMs/transactors, Regression testing, RTL simulation, ModelSim, Synopsys VCS, Pre-silicon emulation, On-board FPGA debug, Test plan development

Performance & Power
PPA trade-offs, Architecture modeling, Design-space exploration, Dynamic power estimation, VCD parsing, Performance-per-watt analysis, Workload and benchmark analysis

Interfaces & Platforms
AXI, PCIe, I2C, LVDS, ARM-based systems, ARM embedded-Linux bring-up, Hardware-software co-simulation

AI / Machine Learning
Neural network accelerators, Machine learning inference, Quantization, Pruning, Benchmarking ML workloads, Video processing

EDUCATION

San Diego State University (SDSU) Dec 2026
Master of Science Computer Engineering
Minors: Computer Architecture, VLSI Design, Digital Systems, Embedded Systems

VR Siddhartha Engineering College
Bachelor of Technology Electronics and Communication Engineering